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INCLUSIVE INFORMATION AND COMMUNICATION STRATEGIES FOR PEOPLE WITH DISABILITY: A LITERATURE REVIEW

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Statement of contributions

This work was completed as part of a student industry placement in the Master of Public Health course at Victoria University. Ms Purva Mahendra Ghandi undertook the literature review under the supervision of Victoria University academic Dr Jerome N Rachele, and City of Melbourne Senior Policy Officer for Access and Inclusion Ms Vickie Feretopoulos. Dr Rachele and Ms Feretopoulos were involved in the conception of the project. Ms Gandhi acquired, analysed, and interpreted the data, and wrote the first draft of this report. Dr Rachele assisted with drafting the report, and was involved in critically revising the content. Both Ms Gandhi and Dr Rachele approved the version to be published.

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EXECUTIVE SUMMARY

This report reviews the literature in the area of inclusive information and communication strategies. It aims to provide recommendations for local governments in Australia, specifically, the City of Melbourne, and how they can be applied to the Melbourne context more broadly. Using the social model of disability, this report highlights the physical, informational and attitudinal barriers that can limit the participation of people with disability, resulting in exclusion; with a specific emphasis on information barriers. Drawing upon theories of universal design, i.e. if a design works well for people with disability it will work well for all, this report further explores the diverse needs of the different groups of disability and their invisible barriers, which can be an impediment to a truly inclusive society.

The key findings of this report are: (i) People with disability are excluded based on decisions made at a societal level; (ii) inclusion and accessibility for all can be achieved only when we understand diversity of disability; and (iii) there needs to be a paradigm shift, whereby equal importance is given to people of all disability types.

Based on the findings of the literature review, recommendations to reduce information barriers and improve communication access for people with disabilities in the City of Melbourne are:

- Improve communication access in a broad range of formats with a focus on written and interpersonal communication.
- Meet the Web Content Accessibility Guidelines (WCAG) 2.0 AA level for people with disability.
- City of Melbourne to partner with local businesses to improve communication access.
- Include people with disability in planning and evaluation of the strategies.
- Promote the use of information and communication technology.
- Establish formal and informal links with the other councils and local government.

1. Introduction

Disability is a complex, multi-dimensional phenomenon. The World Health Organization has defined disability as an “umbrella term” ranging from impairments', restrictions to participation, activity limitations and an interaction between individual (health condition) and contextual factors (environment and personal factors). Disability can therefore be defined as ‘the interaction between persons with impairments and attitudinal and environmental barriers that hinders their full and effective participation in society on an equal basis with others’. Under this definition, 15% of the population globally identify as having a form a disability at some point in their lifetime. Around 18% of Australians, approximately four million people, identify as having a disability (AIHW, 2019).

2. Disability policies in australia

In 1986, disability rights was recognized as a program under the Disability Services Act. It was moved from institutional services to community services, and was followed by the establishment of “public advocates” in most of the States’ Disability Services Act 1986 (DSA) and the Disability Discrimination Act 1992 (DDA). In 2008, Australia ratified the United Nations Convention for the Rights of Persons with Disabilities (people with disability Australia, 2018).

The National Disability Strategy 2010-2020 (Council of Australian Governments, 2011) and United Nations Convention on the Rights of Persons with Disability (UN General Assembly, 2006) each recognise the rights of people with disability to participate on an equal basis to those without a disability, with a focus on health, and maximising social and economic participation. Despite the intentions of these policies and strategies, their aspirations are unlikely to be achieved without inclusive design.

3. Models of disability

There have been two prominent models have been developed in the past 50 years to understand the disability discourse: the medical model and the social model (Haegele, 2016). In the medical model, disability is understood as an individual medical phenomenon and a result of impairment of body function. The solution or treatment is focused towards fixing or eradicating the impairment by placing people with disability under the medical profession in rehabilitation programs. In contrast, the social model of disability, unlike the medical model, breaks the causal link between impairment and disability. In other words, for people with disability, the emphasis for the cause of their economic and social disadvantage shifts from ‘impairment’ to the role of ‘society’ in restricting opportunities to engage in social and economic activities (Barnes et al, 2010). More recently, the World Health Organization has developed a biopsychosocial model which incorporates aspects of both the medical and social models of disability (Figure 1). In the biopsychosocial model, disability is an interaction between features of a person, and features of the overall context in which the person lives. Sometimes almost entirely one or the other.

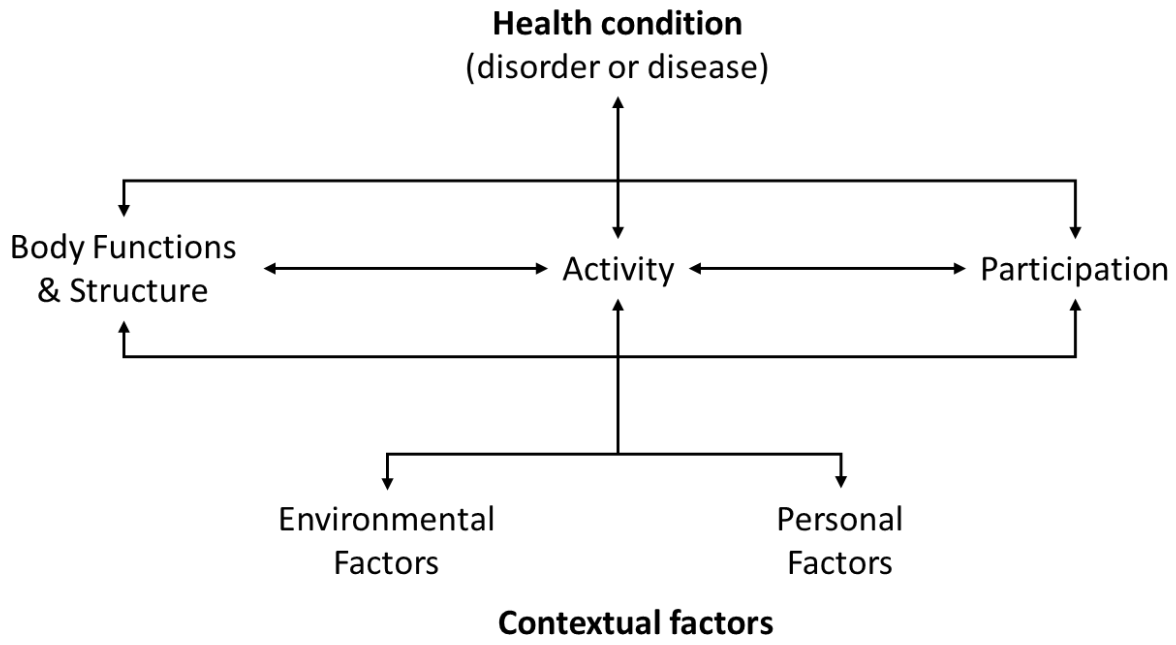

Figure 1 - Biopsychosocial model of disability (Source: WHO International Classification of Functioning)

4. Diversity of Disability

Understanding the different types of disability, and by extension diversity of needs, is an important first step in creating a truly inclusive society (Rachele et al, 2019). Stereotypical views of disability have emphasized wheelchair users or people with vision impairments. Disability is diverse and encompasses invisible and hidden impairments. These include, for example, people with dementia, congenital cerebral palsy, or arthritis (World Bank, 2013). These generalizations are among the major barriers for an inclusive society. Impairment types can be broadly categorized as either physical, sensory, cognitive or intellectual, and psychosocial.

4.1 Physical Impairment

The Australian Bureau of Statistics has estimated that around 14% of Australians have some form of physical impairment (Australian Bureau of Statistics, 2020). Physical impairments can affect a person's ability to move, to get up and down or carry things. Characteristics of the physical environment like steps, roads, footpaths and inaccessible public transport are some of the barriers that can result in limitations in activity participation, such as access goods and services

4.2 Sensory Impairment

This most commonly affects hearing and sight and is associated with ageing. Approximately 80,000 Victorians have vision impairments (Australian Bureau of Statistics). Inaccessible information in the form of standard print and lack of provision of information in audio format are some of the barriers affecting participation their day-to-day lives. Hearing impairments usually affect the ability to communicate or comprehend information, limiting access to

telecommunication and media such as cinema, announcements, television and spoken instructions

4.3 Cognitive/ intellectual impairment

This can result from either acquired brain injury or an existing neurological condition and affects a person's ability to learn and comprehend. People with cognitive or intellectual impairment often face difficulties in coordination, control and communication of their thoughts and actions. Some of the solutions to overcoming barriers include providing information in plain English, or Easy English that uses pictograms and diagrams to convey essential information.

4.4 Psychosocial impairments

One in Five Australians will be affected by a psychosocial impairment at some point in their lives (Australian Bureau of Statistics, 2020). Examples of psychosocial impairment include depression, bipolar, post-traumatic stress disorder, eating disorders and schizophrenia. Difficulties mostly arise from stereotyping and hostile attitudes.

5. Disabling Barriers

Disabling barriers have been classified into three dimensions: physical, informational, and attitudinal.

5.1 Physical Barrier

Physical barriers can be natural or manmade, structural in nature, and can cause mobility restrictions (Centre for Disease Control and Prevention, 2013). These barriers are created by elements of buildings or architecture: building entrances, stairs and hallways, layout of rooms, width of footpaths or inaccessible transport. For example, an information kiosk that is too high for a person using a wheelchair, or poor lighting that restricts a person with low vision (Ontario Human Right Commission, 2013).

5.2 Information and Communication Barrier

Barriers to information and communication can arise while sending or receiving information. These barriers typically affect people with sensory or intellectual disability, or people who use different forms of communication. People with cognitive impairments may find themselves disabled by the use of long sentences, technical jargon and the use of many syllables. Information and communication technologies (ICT) are crucial for people with sensory impairments to participate in day-to-day life. However, these technologies, if inaccessible, can result in barriers in themselves. For example, visual images on websites that do not have a meta tag, meaning that a visually impaired does not receive a text description of the image. ICT access, like physical access, are about human choices rather than technological limitations (Shakespeare, 2013). Some examples of information and communication barriers include use of small fonts, and a lack of large print materials, Braille, screen-reader, closed captioning, or interpreters (Centre for Disease Control and Prevention, 2013).

5.3 Attitudinal Barriers

Attitudes of people without disability are crucial in facilitating inclusion: non-supportive attitudes can trigger and contribute to other barriers. The social model of disability enables us to consider disability as not an individual deficit but instead a social responsibility in which we can make deliberate, considerate and positive choices. These choices can not only lead to an inclusive society but also makes it easier to understand and address the challenges that people with disability face (Centre for Disease Control and Prevention, 2013). Stigma and discrimination towards people with disability varies from interpersonal and institutional, through to policies and laws that result in marginalization that prevent people with disability participating in employment, accessing services and making friends (United Nations, 2018).

6. Universal Design

From a disability policy perspective, universal design has been defined as the design and composition of an environment so that it may be accessed, understood and used: (i) to the greatest possible extent; (ii) in the most independent and natural manner possible; (iii) in the widest possible range of situations; and (iv) without the need for adaptation, modification, assistive devices or specialised solutions, by any persons of any age or size or having any particular physical, sensory, mental health or intellectual ability or disability (Government of Ireland, 2005). If a design works well for people with a disability, it will work for everyone.

Universal design considers usability and aesthetics to be mutually compatible, and that by applying these principles we can achieve inclusion. For example, wider doorways are practical to both wheelchair users and people with prams, and Easy English assists both people with disabilities as well as people who speak English as a second language. Social inclusion refers to attaining an improvement in quality of life by achieving participation of an individual in society and workplace. The outcome of social inclusion is best achieved when designed as accessible for all. Social inclusion based on universal design not only addresses both visible and invisible barriers, but is a comprehensive approach towards the different determinants that result in their exclusion. The prevalence of disability will continue to increase as the population ages, therefore, the implementation universal design will become even more imperative.

7. Why Information And Communication Inclusion?

This report recommends innovations in the area of information and communication inclusion for local government, specifically, the City of Melbourne, and how they can be applied to the Melbourne context more broadly.

The United Nations Convention on the Rights of Persons with Disabilities (UN CRPD) Article 21 - Freedom of expression and opinion, and access to information, has outlined certain measures to uphold the rights to information and communication for the people with disability. Despite most countries globally understanding the need to address physical barriers through standards and legislation, access to information and communication is still relatively a novel concept. There are potential groups of people, including people with neurological disorders, sensory disorders, language learning and literacy disorders, communication disorders, and specific syndromes and birth defects that require information and communication support (Speech Pathology Australia, 2018). According to the Australian

Bureau of Statistics 1.2 million Australians in 2015 had some form of communication impairment, but it is important to note that the number of people requiring support is likely to rise as the population ages: most people with communication needs have conditions associated with aging.

Communication access is not only important for people with disability, but also benefits people from culturally linguistic and diverse populations and individuals with lower literacy rates. Given these 'co-benefits', priority for communication accessibility policies and legislation is warranted. Being a signatory to the UN CRPD, the Australian legislation that is the Disability Discrimination Act has no specific mention of the communication needs or the communication access. However, the National Disability Strategy 2010-2020 in its Outcome 1: Inclusive and Accessible Communities does recognize the need of inclusive communication access in terms of digital technology, but there is lack of recognition of interpersonal needs for communication for people with disability. In the context of the State Disability Action plan that is the Victorian State Disability Plan 2017-2020, there is no mention of information and communication access. The City of Melbourne Action Plan, Melbourne for All strategy, in its Theme 1- Access and Inclusion, does mention the need to eliminate and reduce barriers to enable social inclusion in terms of employment, accessible facilities, accessible information and communication in range of formats, innovation information and communication technologies with an aim to meet Level AA of Web Content Accessibility Guidelines 2.0. The WCAG is an internationally recognised standard created by the World Wide Web Consortium (W3C). These guidelines make content more usable and accessible to people with disabilities and older individuals with changing abilities due to aging (Centre for accessibility,2020). However, there are no other specific strategies or tools for face-to-face, interpersonal or other forms of accessible written and verbal communication.

Notably, despite digital platforms being an accessible form of information and communication for people with disability, this group has a low level of digital inclusion compared to other Australians. The Australian Digital Inclusion Index (ADII) measures the range of digital inclusion in Australia across themes of access, affordability and digital literacy, where higher scores mean higher levels of inclusion (Australian Digital Index Inclusion, 2015). The index score for people with disability is 52.0, 9.9 points below the national average (Australian Digital Inclusion Index, 2019). And while digital inclusion is higher among some groups (e.g. people with sensory impairment show a very high level of access and digital ability compared to the national average) these are often mitigated by low levels of affordability (Australian Digital Inclusion Index, 2018). Moreover, the prevalence of disability increases as age increases, but so does the digital divide. Those aged 65+ are the most digitally excluded age group in Australia with a score of 14.2 below the national average. These findings show that, even though the digital platform is promising and reachable to a larger population, there is a need to investigate effective strategies for inclusive traditional forms of communication like interpersonal, face-to face, written for the population groups that are digitally excluded.

7.1 Information and Communication Technology

Despite people with disability being amongst the most digitally excluded groups, people with sensory impairment, especially those with hearing impairments have high digital ability

(Australian Digital Inclusion Index, 2019). The empowering role of computer technology, learning about new technology and access to the internet were some of the affirmative reasons for the use the ICTs among this group. In comparison to the general population, people with disability rely more heavily on the internet for a range of day to day activities including making video calls, purchasing and selling products, internet banking and also browsing the internet (Australian Digital Inclusion Index, 2019). The Australian Digital Inclusion Index Supplementary Survey for the Deaf or Hard for Hearing community also showed that people with hearing impairments are more likely to maintain multiple internet access points with both fixed and the mobile internet in comparison to the general population, in order to facilitate intensive day-to-day use. Though access and ability are relatively high for the people with disability, affordability, especially in Sydney, Melbourne and Brisbane are among the primary causes for their digital exclusion (Australian Digital Inclusion Index, 2018). Information and communication, when accessible and available, is an important enabler of social inclusion. ICT has the power of bring people together, but at the same time it can also leave people behind if not accessible and affordable.

A United Nations toolkit states specific principles with respect to ICT and disability (United Nations, 2016). First, ICT should be designed to be inclusive to everyone, as opposed to specifically designed for people with disability. And second, it should facilitate information and communication in different cultural, educational, and professional communications. Article 9 of the UN CRPD explicitly mentions the rights for people with disabilities to access information and communication technologies on an equal basis and without discrimination. The Convention defines “Communication” as “including all possible means of communication that may eliminate barriers; the term includes languages, display of text, braille, tactile communication, large print, accessible multimedia as well as written, audio, plain-language, human-reader and augmentative and alternative modes, means and formats of communication, including accessible information and communication technologies”. The UN CRPD therefore calls on Member States to encourage the private sector to deliver accessible services by provision of reasonable accommodation. In order for Member States to comply with the UN CRPD, ICTs must accommodate the different types of disability, as different groups face different challenges and therefore require different strategies. For example, people with vision impairments may need audio or tactile output, while people with hearing impairments may need graphics and texts for information and communication.

Accessible communication also includes assistive technologies that can be used for the interaction of the user with the ICT. These include: input devices for people with reduced mobility, screen readers for persons with vision impairment, supplement for the natural speech (augmentative), non-speech (alternative) communication strategies for people who are speech and writing impaired, and icon-based interfaces for people with cognitive impairment. For these assistive technologies to be user friendly for people with disability, service providers need to be trained and supported, and require a significant amount of literacy around diversity of disability and their specific needs. These assistive technologies can be provided at a specific location within these facilities, for example: schools, local council offices, rehabilitation centres, or as a part of workplace accommodation services. Therefore, while assistive technologies are an effective strategy for a wide spectrum, mobile applications address a variety of daily living needs that can be downloaded and used.

In a study by Barlott et al (2019), participants with intellectual disabilities reported that information and communication technologies, specifically handheld mobile applications "open the door to possibilities". The study found that mobile phones were the most common device for accessing information and communication technology, which is in line with current ICT trends: increased use of mobile phones and smartphones with declining desktop use. Mobile phones thus have potential to reduce the digital divide and the websites of essential service providers and governments need to be easy to navigate and accessible for the mobile-only users.

8. Inclusive Strategies Used Globally For Information And Communication

8.1 Examples Of Strategies To Overcome Interpersonal Communication Barriers

There is a need to increase practices, tools and strategies regarding information and communication. Specifically, interpersonal and accessible written formats, as access to communication is limited, with access mostly focusing on physical access or digital technologies. There remains a considerable gap in research and evaluation to determine which approaches and strategies are best and most sustainable for the broadest group of people with specific communication needs. Communication support needs are defined by Law et al. 2007 as "difficulties associated with one or more aspects of communication. This includes verbal understanding, expressive language, speech and the capacity to understand someone's intended meaning rather than the words themselves. It also refers to literacy and other means by which individuals interact with one another". The National Disability Strategy 2010-2020 includes inclusive and accessible communication policy that is relevant to local governments. Local governments play an important role in digital information and communication, ensuring that government websites are complaint with the Web Conent Accessibility Guidelines, and providing alternative formats of documents in Easy English, Braille and Auslan. Local governments are also required to investigate alternative communication strategies and tools to overcome the communication barriers (Disability Inclusion Planning, A Plan for Local Councils, 2016). Some examples of strategies to overcome barriers for people with communication difficulties include:

- The Ku-Ring Gai City Council in NSW provided inclusion support training to its staff working with children who have disabilities that affect their communication, following the Disability Inclusion Planning Guide for Local Governments. Providing training not only empowered staff with knowledge, but also gave them the confidence to communicate appropriately with children.
- The City of Marrandurah in WA, as a part of their inclusion initiative, provided portable hearing augmentation devices for council events to support people with hearing loss.
- The Queensland Government has designed a booklet, 'Way With Words', with the purpose of promoting inclusion and the fair portrayal of people with disability. The intention of the booklet is to aid communicators to use appropriate language and techniques while communicating with people who require communication assistance.

These are just some examples of some appropriate practices that can be effective and adopted as strategies for people with disabilities to aid communication access. There is strong evidence for the use of alternative and augmentative forms of communication for people who need communication access. However, there is limited literature that focuses on the interpersonal, face-to-face mode of communication that aims to train staff on capacity building as a strategy for inclusive communication. The Communication and Inclusion Resource Centre, SCOPE, in collaboration with people with communication difficulties, the Communication Access Network (CAN), peak bodies and key organizations are promoting the concept of communication access, with the vision of creating communities that are inclusive and accessible with respect to all people including communication difficulties, physical disabilities, reading difficulties, vision impairments, hearing impairment and intellectual disability. SCOPE assesses businesses and services with a checklist to determine if they meets the minimum standards to be awarded the Communication Access symbol. The minimum standards include staff that are willing to use the different forms of communication, providing an environment that supports communication, signs that are clear, and written information that is accessible (Communication Access for All, 2015)

9. Key Takeaways From This Report

- The transition from medical model to social model of disability is strongly related to accessibility, and disability should be viewed as a result of an inaccessible environment.
- Despite impairment resulting in restrictions for individuals, disability is created by environmental hurdles, hostile attitudes and social barriers.
- Social inclusion and accessibility is possible only by understanding the diversity of disability and not generalizing around the term.
- Social inclusion can be achieved when it considers all people, rather than just people with disability. Further, social inclusion with universal design principles addresses both the visible and the invisible barriers leading to disability.
- Co-designing or involving people with disabilities in planning and evaluation of strategies and actions is a key to successful action plans, as “nothing for us, without us”.
- Access to inclusive information and communication are as important as physical access. Inaccessible communication can result in social exclusion and limited opportunities for a broad group of people with diverse disabilities.
- Inclusive communication not only benefits people with disability but also supports the communication needs of culturally linguistic and diverse populations, including individuals who do not use English as their first language and people with lower literacy levels.
- The prevalence of disability increases as the age increases but so does the digital divide. Those aged 65 years and older are the most digitally excluded age group in Australia. These findings show that even though the digital platform is promising and reachable to a large population, there is a need to investigate effective strategies for inclusive traditional forms of communication like interpersonal, face-to face, and written for the population groups that are digitally excluded.

- People with disabilities greatly rely on the internet, especially their mobile phones for their day-to-day activities in comparison to the general population. The digital divide in metropolitan cities is strongly related to affordability, rather than ability and access.
- For any organization, govnerment or service to fulfill the communication standards, it is essential to train staff with the help of communication training partners about the diversity of disability, in order to facilitate inclusion for a broad range of individuals with communication needs.
- As with legislative support to physical access in the form of building codes and public transport standards, for sustainable change in communication access, there is a need to increase support through legislation.
- There is appropriate emphasis in action plans for websites to be accessible, but communication accessibility should be supported for a broad range of contexts including face-to-face, written and telephone services with evidence-based strategies.
- It is important that government communicators follow guidelines for accessible communication, such that everyone has the same access and opportunities to attend public events, read publications, use websites, and access information about policies and programs.

10. Recommendations

The following are a series of recommendations that, if not already being undertaken, the City of Melbourne should consider adopting to reduce barriers to communication for people with disability.

Improve communication access in the City of Melbourne in a broad range of formats and diverse disabilities such as interpersonal and written communication

- Train the council staff and communicators in alternative forms of communication to facilitate inclusion for the different groups of people with disability.
- Formulate a standard document (i.e. guideline or toolkit) to communicate in alternative forms including interpersonal, written, telephonic and information on the web, providing detailed information in Easy English, Auslan, closed captioning and formatting documents for easier perception of information.
- Promote the use of augmentative and alternative forms of communication, while using services and perceiving information.
- Gain accreditation as a Communication Accessible council, by collaborating with communication partners like SCOPE. This will not only aid in accessibility and inclusion, but also empower staff and boost their confidence while communicating with people who require communication support.

Ensure that the council websites meet the Web Content Accessibility Guidelines (WCAG) 2.0 AA level for people with disability.

- Audit the existing website regularly to check if it is meeting the WCAG guidelines.
- Train the Council IT staff in WCAG to maintain the website and a web accessibility consultancy to help design and deliver the website to the required standards

Establish formal or informal networks with other councils that will facilitate the sharing of information on inclusive practices. These networks will expose council staff to different ideas, initiatives and also encourage collaboration especially where community needs are similar. Moreover, it will aid in raising inclusion by valuable points of comparison for inclusive strategies

Promote the use of information and communication technologies for people with disabilities

- Adopt best practices for information and communication technologies (The ICT Opportunity for a Disability-Inclusive Development Framework, 2013)
- Strengthen research and develop affordable ICT-enabled solutions for people with disabilities.
- Incorporate accessibility guidelines in procurement policies
- Update the disability legislation, to include ICT in the definitions of accessibility

Advocate strongly for communication access in different formats along with digital standards in the future disability action plan

The City of Melbourne should partner with local businesses to facilitate communication access for people with disability

- Support and encourage businesses to implement initiatives that focus on disability inclusion. For example, The City of Melville in Western Australia has collaborated with shopping centers to produce a of booklet that includes tips and actions on how they can implement inclusive practices such as color contrast, clear signage, customer services and communication

Include people with disability in planning and evaluating inclusive practices.

- The Inclusion of people with disability in the form of an advisory Committee is one of the most effective practices. This not only provides local government with access to feedback and insight, but also empowers people with disability in decision-making and gives them a voice within local government (Disability inclusion planning: a guide for local Government 2016).
- Implement mobile applications like 'Snap Send Solve'. Such applications enable members of the community to communicate with councils directly as they come across barriers such as signage accessibility in public spaces. The person facing difficulties can take a picture and act on the spot by sending it to the council. These strategies are used by many councils across Melbourne and can result in direct engagement with the community

11. Conclusion

Inclusion of people with disabilities in the different aspects of society is one of the many challenges of the global development agenda. The City of Melbourne strongly advocates access and inclusion in its current disability action plan in terms of digital accessibility. However, there is a need for the council to consider other forms of communication to achieve information and communication inclusion.

The recommendations outlined in the report seek to improve the current communication practices and aim to move one step closer to be an information and communication inclusive city for people with disabilities.

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